

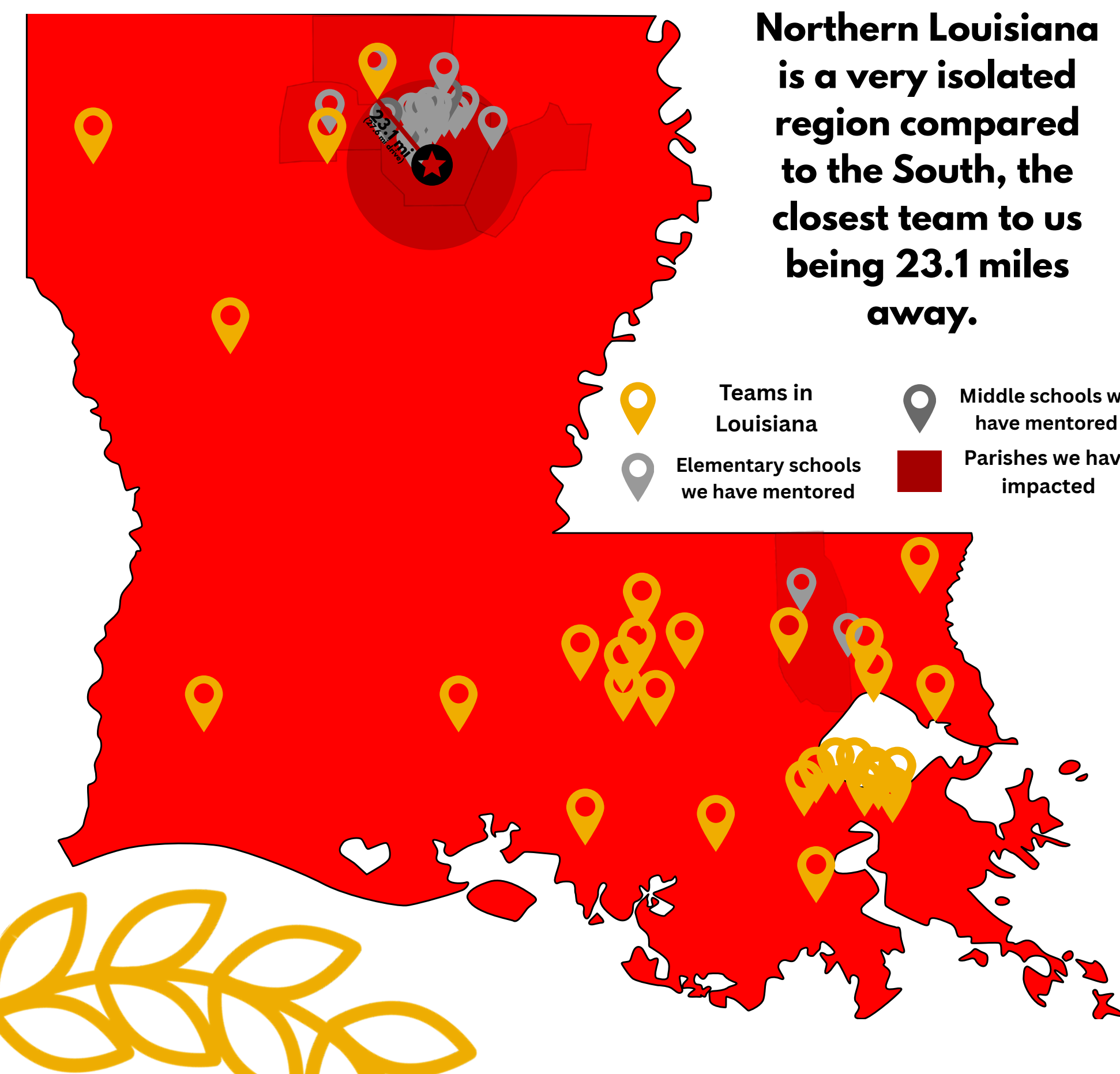
MAGNATECH

#3468



"AN ABUNDANCE OF CHARACTER"

In a community where brilliant minds often lack the tools to succeed, we provide the high-tech bridge to a STEM career. Our focus isn't on beating other teams, but on ensuring our members are more skilled, more capable, and more confident than they were a year ago.



Over the
last 3 years

...

45 total events
2 teams created
6 Camps
31 Schools impacted
~1,769 Outreach hours

Students deserve an opportunity to learn and flourish in an environment that supports them!

MAGNATECH

#3468

REBEL ROBOTICS



JOIN US TO CELEBRATE OUR TEAM'S ACCOMPLISHMENTS AND
DISCOVER THE FUTURE GOALS OF MAGNATECH!

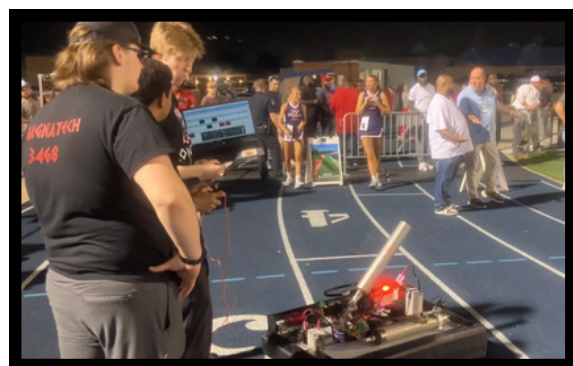
ROBOTICS SHOWCASE BANQUET
THURSDAY, MAY 14TH
6 - 8 PM

WEST MONROE HIGH SCHOOL - EDGEWOOD BUILDING

LOCATED ACROSS FROM THE AUDITORIUM ON TRENTON STREET SIDE OF CAMPUS



COMPETITION SCOUTING APP DESIGNED & LAUNCHED



T-SHIRT CANNON COMING TO REBEL STADIUM THIS FALL!



STARTED AND MENTORED 2 ELEMENTARY FIRST FLL TEAMS



BAYOU REGIONAL TEAM ATTRIBUTE AWARD: IMAGERY

PARTNERED WITH:

SPECIAL THANKS TO OUR SPONSORS!

HALLIBURTON

INTUITIVE
Foundation

ATMOS
energy.

Goldman
EQUIPMENT

CONCEPT
INDUSTRIAL SYSTEMS



A&H GAMES
Games • Cards • Electronics

drax



MAGNATECH

#3468

REBEL ROBOTICS



In the Greek Language,

MAGNA = great
TECH = technology

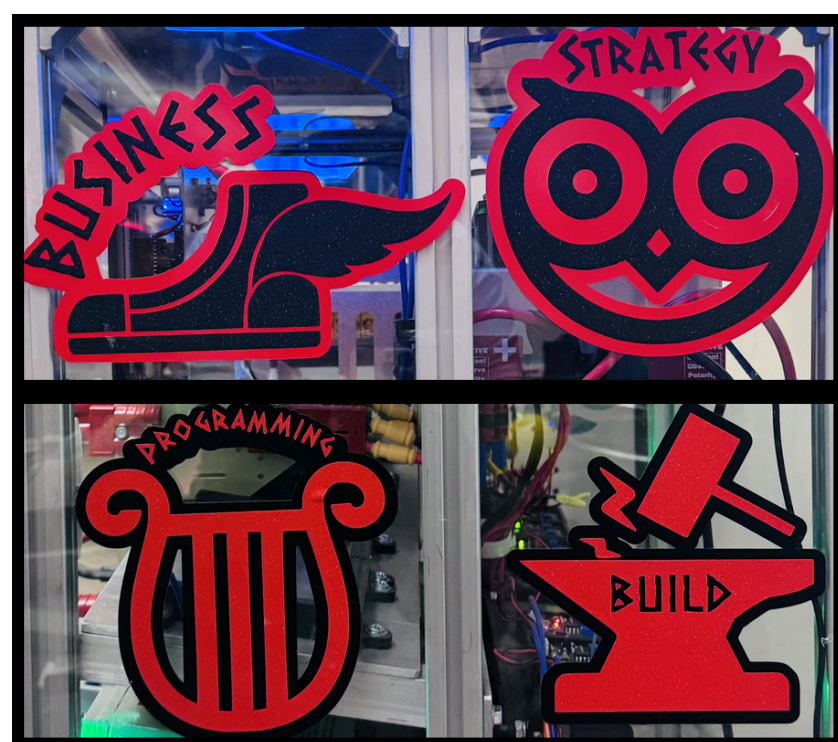
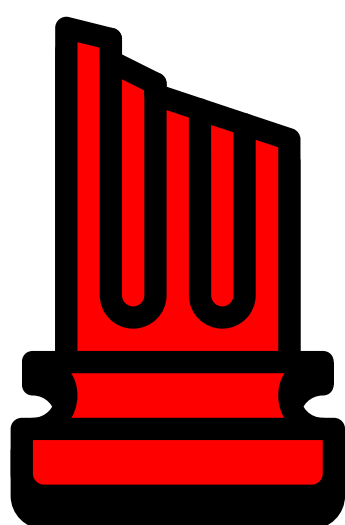
JOIN OUR TEAM!

APPLICATIONS DUE

MAY 14TH

TEAM MOTTO: "SCIENTIA POTENTIA EST"

TEAM VISION: OUR TEAM WISHES TO INSPIRE AND MOTIVATE OUR MEMBERS TO GAIN CONFIDENCE IN THEIR LEADERSHIP AND TECHNICAL SKILLS THROUGH THE ENGINEERING DESIGN PROCESS TO ENSURE THEIR FUTURE CHALLENGING ENDEAVORS ARE UNDERTAKEN WITH GRACE AND EXCITEMENT.



MAGNATECH

#3468

REBEL ROBOTICS



You are invited to go behind the scenes to learn more about our team. Join us for an exclusive look at our mission in motion. Tour our build shop, meet team subgroup leaders, and **check out our practice field to see the Bayou Regional FRC Imagery Awarded Robot: Kronos 2.0** in action.

SEE THE IMPACT. HELP US BUILD WHAT'S NEXT.

SAVE THE DATE

COMMUNITY OPEN HOUSE

TUESDAY, SEPTEMBER 29TH

5:30 - 7 PM

WEST MONROE HIGH SCHOOL - WING 7 MAGNATECH

LOCATED BETWEEN THE MAIN OFFICE AND THE AG BUILDING ON THE RIGGS STREET SIDE OF CAMPUS



Our team's mission is to close the gap between local brilliance and the rising demand for high-tech careers. We serve as a vital bridge for a community full of potential.

We strive to win the 'long game' by investing in the development of our team members. By providing accessible STEM activities at community events to ensure our community is geared with the skills, confidence, and knowledge. At our core, we believe that empowering the next generation of leaders and innovators is the ultimate win.



Teams in
Louisiana



Elementary schools
we have mentored



Middle schools we
have mentored



Parishes we have
impacted

THE CHALLENGE

Economic Barrier: 26.3% Poverty Rate in West Monroe creates an accessibility gap in families.

Academic Gap: Only 8% math/science proficiency currently recorded among local students.

Industry Demand: STEM jobs are the fastest growing sector in Louisiana.

THE OPPORTUNITY

Student Drive: 51% of students in Louisiana express high interest in STEM.

State Progress: Louisiana's education ranking jumped from 46th to 37th in just 4 years.

Proven Success: 91.2% of our members are interested in having a STEM future.

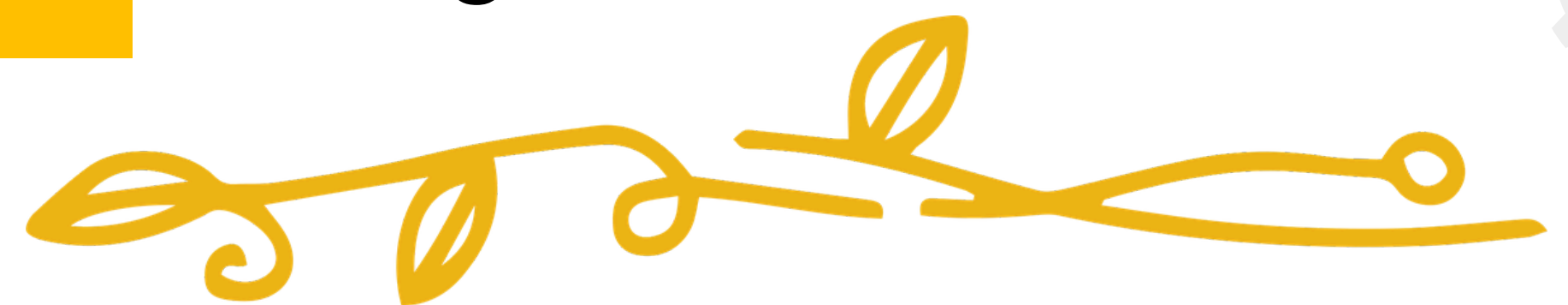
OUR RESPONSE

**Total Reach:
31 Schools across
5 Parishes**

**Total Effort:
~1,769 Outreach
Hours**

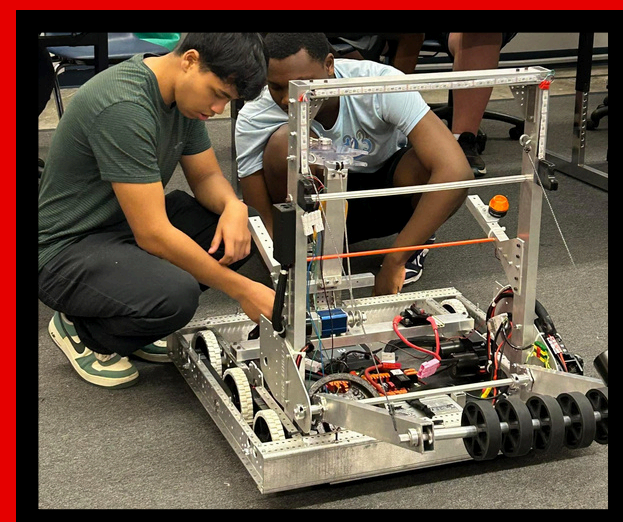
**Direct Mentorship:
176 Students / 2 FLL
Teams**

**STEM Pipeline: ~30
Camper Retention
for Summer Camps
and 3 current team
members**



FRC Shop Tour and Workshop

We hosted and mentored Union Parish's Farmbots on soldering and wiring, and gifted the team a Kit bot.



City Council Meeting

We demonstrated our robot to the City of West Monroe to advocate for our team.

FLL Mondays

We started and mentored 2 FLL teams, 33825 challenge & 72405 explore, and hosted the teams every Monday to teach them how to code, CAD, and have experience with STEM.

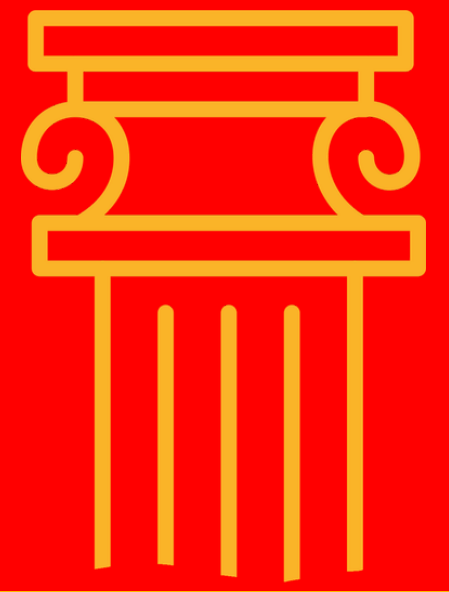


Summer Camps

Hosted camps for elementary and middle schoolers to explore 3D printing and engaging STEM activities

MAGNATECH

#3468



"AN ABUNDANCE OF CHARACTER"

In the Greek Language,

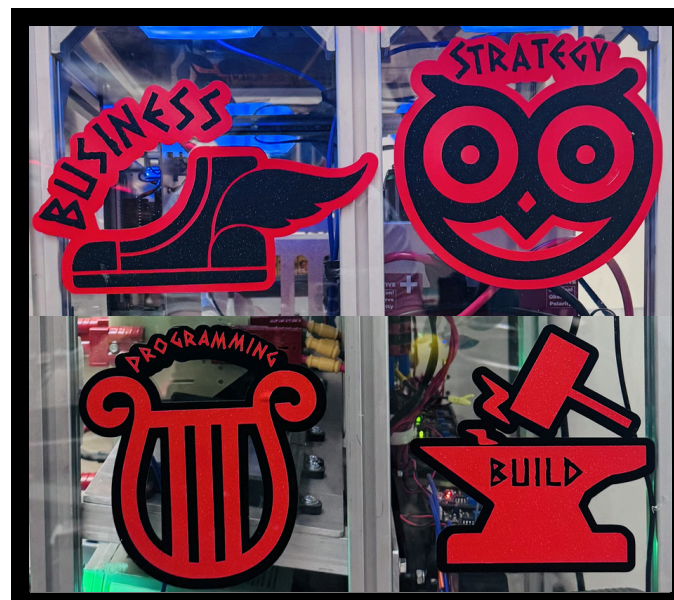
MAGNA = great
TECH = technology

MAGNAtch was born in the Greek Language class at West Monroe High School 2011.

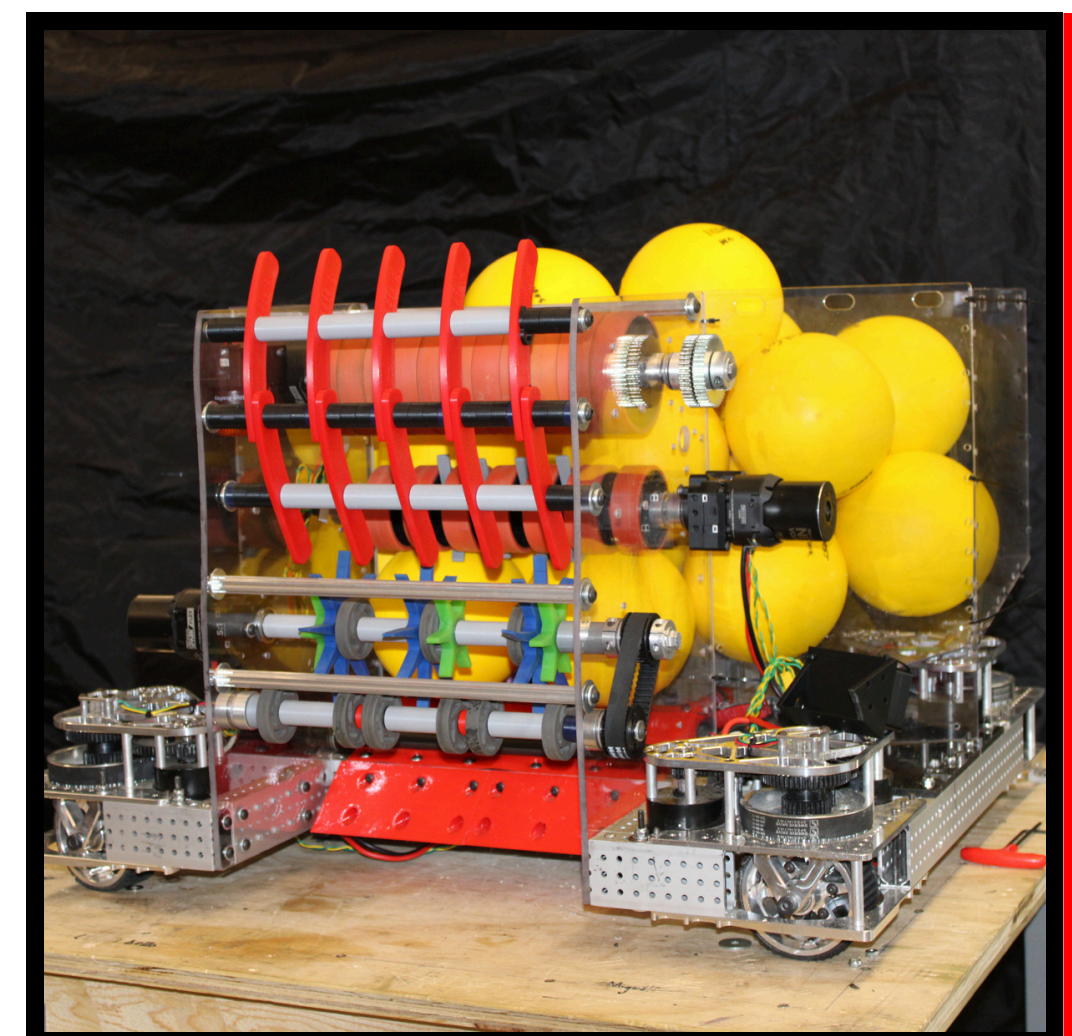
This Greek origin is was influences our every action and look of our team today!



Our custom-mapped controls are reconfigured annually for peak efficiency, styled with our signature red and iconic Greek columns.



Our cart is adorned with both imagery of our team, being laural leaves and columns, and the greek gods that represent each of our subgroups.



Every year, we choose a greek figure to name our robot after. This year we chose Kronos, Titan of Time. This name is put onto the robot to further represent our team.





The team designs a t-shirt design every year, incorporating the Greek God we chose and the game theme.



This year, we decided to design jerseys to further illustrate our team's motif.



Along with the jerseys, we decided to design and give bandanas to leadership. They have our team numbers and leadership role.



The staff is of the symbol of Kronos, Titan of Time, and it's purpose is to further focus on our theme.



For every competition, we design new buttons, keychains, earrings, and bracelets to give away at competitions. They always have Greek Symbolism like laurel leaves and columns, along with our team colors.

At MAGNAtch, brilliance is only the foundation. We strive to be the 'living pillars' of our community. Our team works in harmony to elevate each other and demonstrate an “Abundance of Character” that shines through every uniform, project, and interaction!



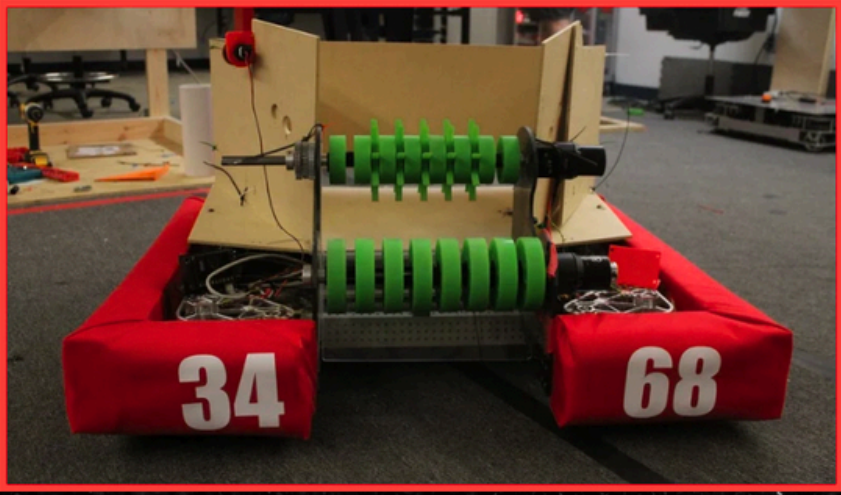
MAGNATECH

#3468



"AN ABUNDANCE OF CHARACTER"

KRONOS 3468 HP 



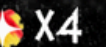
GREEK ROBOT. FRAME: 25 X 30", WEIGHT: 100 LBS.

Ability AETHER

KRONO'S LIGHT WEIGHT FRAME ALLOWS THE ROBOT TO GLIDE THROUGH TRENCHES AND SOAR OVER BUMPS

  **FUEL BARAGE**

GIVES THE ROBOT THE ABILITY TO DISPURSE FUEL IN NOT ONLY THE HUB BUT ACROSS THE FIELD

OFFENSE  X3 DEFENSE  X4 EST. FEB. 21ST

 @frcmagnatech3468  @magnatech_3468

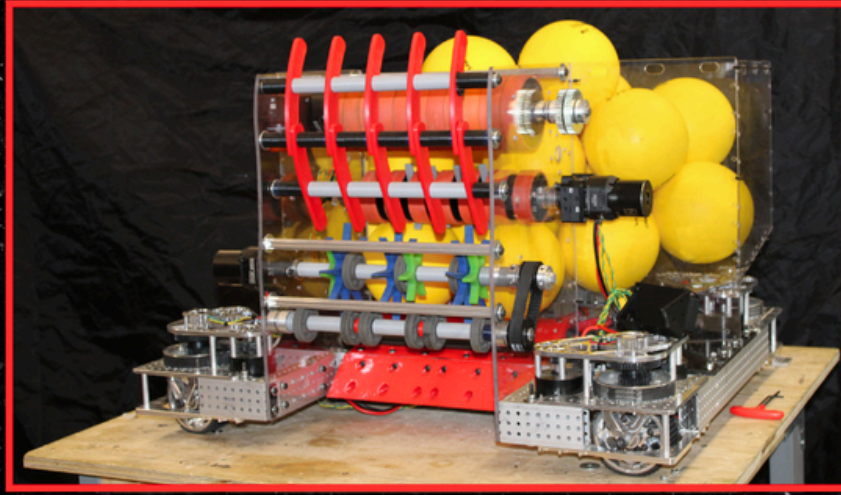
 @frc3468.MAGNatech  @MAGNatech 3468 Robotics



Arkansas Regional =
KRONOS 2.0

Our robot did not work the way we intended. The drive team had to switch to a defense strategy, and upon returning from AR, the team decided to build a modified EVERYBOT shooter/intake to increase efficiency.

KRONOS 2.0 3468 HP 



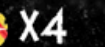
GREEK ROBOT. FRAME: 25 X 30", WEIGHT: 100 LBS.

Ability AETHER

KRONO'S LIGHT WEIGHT FRAME ALLOWS THE ROBOT TO GLIDE THROUGH TRENCHES AND SOAR OVER BUMPS

  **FUEL BARAGE**

GIVES THE ROBOT THE ABILITY TO DISPURSE FUEL IN NOT ONLY THE HUB BUT ACROSS THE FIELD

OFFENSE  X3 DEFENSE  X4 EST. MAR. 29TH

 @frcmagnatech3468  @magnatech_3468

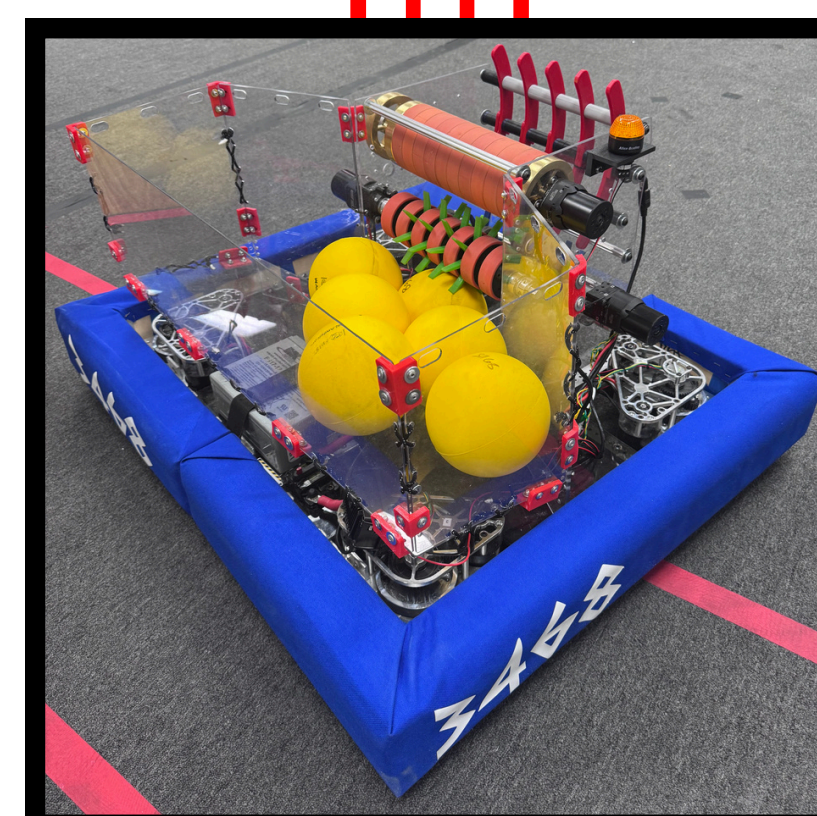
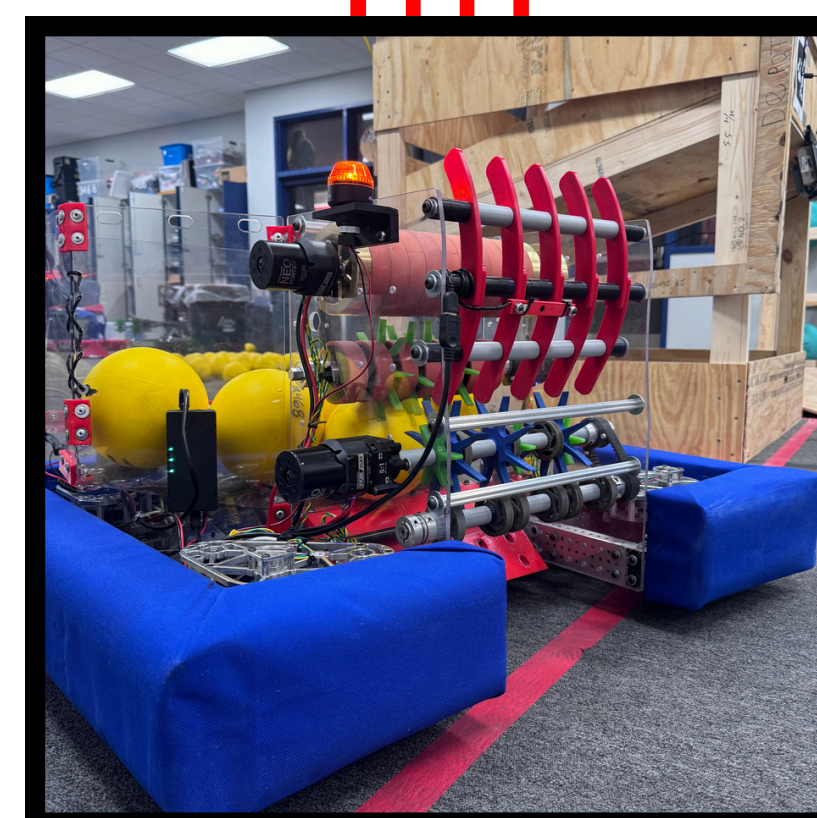
 @frc3468.MAGNatech  @MAGNatech 3468 Robotics

Shooter/Intake

Our intake system is a dual subsystem that works as an intake and a shooter. The lower wheels grip the fuel and push it back into the hopper. When ready to score, the upper wheel then shoot the fuel back.

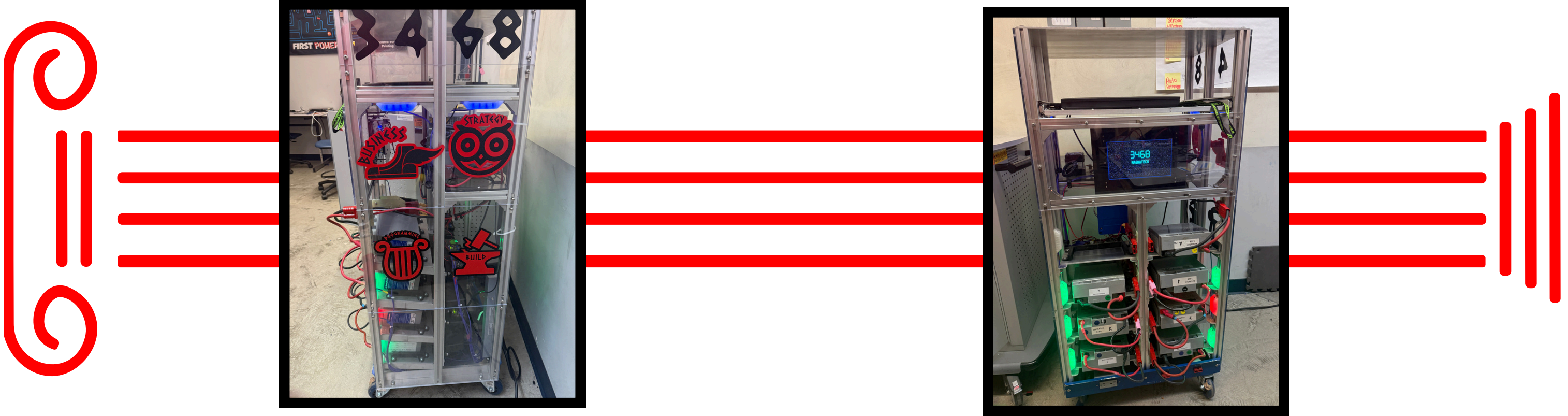
Fixed Hopper

We have a in frame built hopper that it strong and flexibly enough to withstand anything thrown at us.



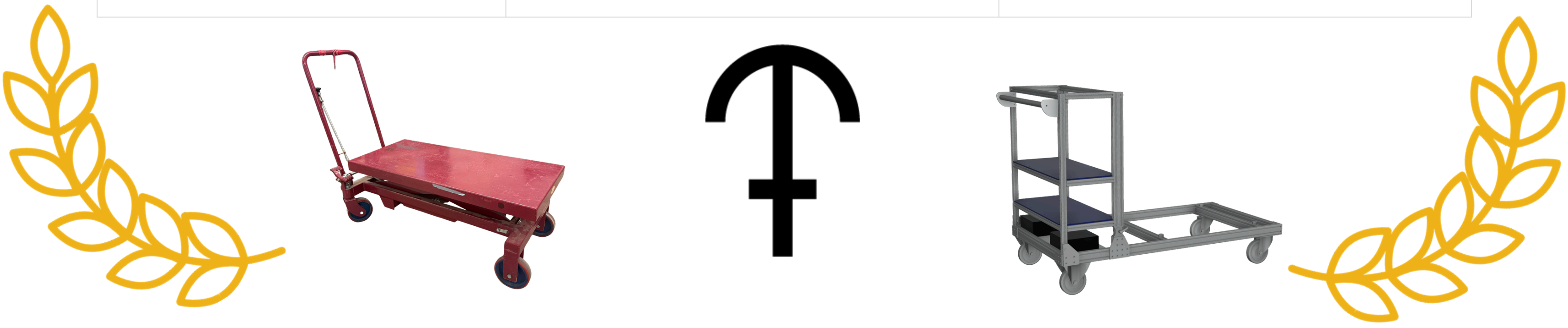
INTEGRATED BATTERY MANAGEMENT & DIAGNOSTIC STATION

Feature	Engineering Function	Team Impact
8-Port Charging Bay	Rapidly charges a full fleet of batteries simultaneously.	Readiness: Guarantees a fresh, 100% charged battery is available for every back-to-back match.
Computerized Diagnostic Port	Integrates a battery tester with a top-mounted computer system.	Reliability: Allows students to record and analyze discharge data to "retire" old or defective batteries.
Integrated Station Link	Provides a dedicated power outlet to charge the Universal Drive Station.	Continuity: Ensures the driver's laptop never loses power between matches or during tethering.
Greek-Themed Branding	Finished with custom Greek-themed vinyl wraps.	Imagery: Transforms a piece of utility equipment into a professional part of the team's visual brand.



A "NEW ERA" CUSTOM ROBOT CART

Feature	Engineering Benefit	Operational Impact
Modular Rail System	High adjustability for varying robot frame perimeters.	Longevity: Cart remains viable across multiple game seasons.
Integrated Docking	Secure mount for the Driver Station laptop and controllers.	Risk Mitigation: Prevents damage to critical control electronics.
Foldable Chassis	Collapsible sections for transport and pit storage.	Efficiency: Optimizes limited 10'x10' pit space.
Onboard Power Hub	Integrated charging ports and 3-prong outlets.	Readiness: Ensures 100% uptime for scouting and drive tablets.
Greek-Themed Imagery	Custom vinyl wraps and synced Alliance-color LEDs.	Branding: Creates a cohesive, high-end "Imagery" package.



MAGNATECH

#3468



"AN ABUNDANCE OF CHARACTER"

COMPSONOCRATES.COM

CompSocrates is a “student-prompted” website, coded by AI, for the purpose of scouting other robots.

TRANSPARENCY

Practice Scouting: with this data, we’ve created a Practice Scouting page where a match is pulled, and a Match Scout can scout all 3 robots on an alliance, then get an accuracy. There’s a mode where the video can be paused, and a mode where it can’t.

Flags: there is a flagging system where when a Match Scout reports questionable* data, it is flagged, but not excluded in any way (so that if it’s a false alarm good data doesn’t get excluded).

Alliance Accuracy: by pulling TBA data, we can see the accuracy of alliance scores calculated by Match Scouts, and estimate individual Match Scout accuracies.

Accuracy Verification: experienced scouts verify correctness of rookie scout data by rescouting matches and gathering their own data.

*Questionable data is currently any of the following: no teleop score, excessive auto score, and excessive human player score.

SCOUTING

There are 7 different forms that are made to be filled out for different purposes:

Match Scout Form: determine a specific team’s score in a match.

Lead Scout Form: rate the skill of and determine the strategy of a whole alliance and its individual teams.

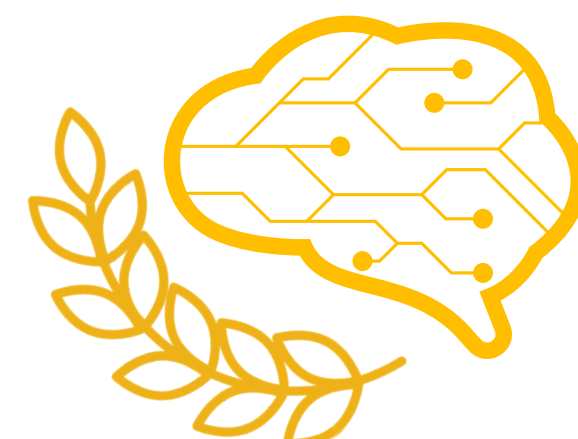
Pit Scout Form: gather general information about every team’s robot.

Team Strategy Form: gather match preferences of every team’s robot.

Match Strategy Form: determine the role of each team for a match.

Drive Reflection Form: determine what each team actually did for a match.

Helper Form: log any time pit members go to assist another pit.



STRATEGY

There is an analytics page that holds all of the form data. The first seven pages are all spreadsheet-esque and hold the raw form data. The next five present that data: Team Averages: the average number of points scored for each team.

Match Breakdown: shows what each team scored in a match.

Rankings: puts each team in ranks based on their average and highest scores.

Team Breakdown: averages every scoring objective, pairs it with Statbotics data, and gives each team a page.

Pick List: sorts teams by official rank, and shows these numbers: Official Rank, Scouted Rank, Scouted Average, Scouted High, Official EPA.

This is a picture of the top of the REBUILT Match Scout Form.



THE FUTURE

Although we didn't want to deploy this app due to it's production beginning in January, we have code for other teams to use this app. Event Selection is a prime example of this, and regardless of event your matches are synced across the website correctly. There's also code stored for a Form Builder where coaches can build their own scout forms for each event.

EASY

Assigning matches to scouts for the day is as easy as a few buttons. There are even options to decide when scouts swap out, and a priority team list if you have less than six scouts.

INFORMATION		PRE-MATCH		AUTONOMOUS				
INFORMATION		PRE-MATCH		STATS			FUEL	
MATCH †	TEAM †	SCOUT †	STARTING POSITION †	PRELOAD †	BPS †	CARRY †	FUEL †	HUMAN PLAYER †
P1	4522	Finn Primeaux	-	7-8	10-13	43-53	~8	0
P1	3937	Caden Floyd	Middle	5-6	0	33-42	6	6
P1	11102	Caden Floyd	Middle	0	0	0	5	5
P1	8753	Brandon Villanueva	Middle	7-8	4-6	13-23	0	0
P1	4764	Adrian Mckoin	-	3-4	0	0	0	0

This is a picture of some of the Match Analytics for Arkansas Regional.

